Mahdi Kallel

PhD candidate in reinforcement learning

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Google Scholar

Profile

- Graduating from the University of Würzburg in April 2027, advised by Carlo D'Eramo. I started out designing algorithms for exploration and policy optimization, and now study the learning dynamics of neural networks, in particular memory and plasticity in continual learning, and how reinforcement learning changes them compared with supervised learning.

Publications

The Spectral Anatomy of Plasticity Loss 2026

Uses random matrix theory to explain loss of plasticity. Over a stream of tasks, the directions that hold earlier knowledge absorb a growing share of each new gradient. Making them read-only keeps the network trainable without losing what it learned.

Mahdi Kallel, Nour Jamoussi, Johannes Tölle, Marios Kountouris, Carlo D'Eramo

www.eurecom.fr/en/publication/8988 (AXIOM Workshop at NeurIPS 2026)

Do Not Imitate, Reinforce: Iterative Classification via Belief Refinement 2026

Trains classifiers with reinforcement learning instead of imitating labels with cross-entropy: a recurrent agent refines its prediction step by step and is rewarded for each improvement. It achieves a better balance of accuracy and calibration than supervised training, and stays calibrated under human label noise, where cross-entropy grows overconfident.

Mahdi Kallel, Johannes Tölle, Ahmed Hendawy, Carlo D'Eramo

arxiv.org/abs/2604.22110 (Transactions on Machine Learning Research (TMLR), 2026)

PPO+: Enhancing Proximal Policy Optimization 2026

Shows that PPO is held back by its critic. Training the critic on off-policy data, while keeping the policy update on-policy, substantially improves PPO and narrows the gap to off-policy algorithms, without implementation tricks.

Mahdi Kallel, Jose-Luis Holgado-Alvarez, Samuele Tosatto, Carlo D'Eramo

rlj.cs.umass.edu/2026/papers/Paper8.html (Reinforcement Learning Journal (RLC 2026))

Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning 2026

Ahmed Hendawy, Henrik Metternich, Théo Vincent, **Mahdi Kallel**, Jan Peters, Carlo D'Eramo

arxiv.org/abs/2510.02590 (ICLR 2026)

Augmented Bayesian Policy Search 2024

Makes Bayesian policy search aware that it is solving an MDP: the performance difference lemma turns a learned critic into the prior over returns. The method outperforms earlier Bayesian optimization approaches, with the largest gains on high-dimensional policies.

Mahdi Kallel, Debabrota Basu, Riad Akrour, Carlo D'Eramo

proceedings.iclr.cc/paper_files/paper/2024/hash/7eca17ef54789b0663cab421f2e9dbf5-Abstract-Conference.html (ICLR 2024)

Cyclicity-Regularized Coordination Graphs 2024

Oliver Järnefelt, **Mahdi Kallel**, Carlo D'Eramo

rlj.cs.umass.edu/2024/papers/Paper40.html (RLC 2024)

Research Experience

INRIA Lille, SCOOL Research Group, Research Intern — France, 2022
- Started the work on Bayesian policy search with deterministic policies that became Augmented Bayesian Policy Search (ICLR 2024).

EURECOM, Research Intern — Biot, France, 2020
- Compared centralized and decentralized approaches to mobile traffic prediction and load balancing.

Industry Experience

Mcovision (independent), Machine Learning Engineer — Paris, France, 2021
- Built a classifier of echocardiography videos for cardiac amyloidosis and other rare heart diseases with Hôpital Cochin, from data cleaning to training.

SustainEcho, Research Intern — Paris, France, 2021
- Explored few-shot and hierarchical learning methods for classification with little data.

Education

PhD **University of Würzburg**, Reinforcement Learning — Germany, 2023 – 2027
- Expected April 2027. Supervisor: Prof. Carlo D'Eramo.
- Teaching: Introduction to Machine Learning; Introduction to Reinforcement Learning.

MSc **Université Paris-Saclay**, Optimization — France, 2021 – 2022
- Coursework: Convex analysis, compressed sensing, dynamic programming, tropical algebra, game theory, stochastic optimization and approximation.

MSc **ENS Paris-Saclay**, Mathematics, Vision, Learning (MVA) — France, 2020 – 2021
- Coursework: Convex optimization, reinforcement learning, kernel methods, computer vision, advanced deep learning, machine learning for text and graphs.

BSc **Télécom Paris / EURECOM**, Data Science — France, 2018 – 2020
- Coursework: Advanced statistical inference, statistical signal processing, machine learning, database management systems, cloud computing and security.

Supervision, Talks and Service

Supervision: Johannes Tölle, MSc thesis, University of Würzburg (2026). The thesis received the VDE Bayern Award 2026, and Johannes co-authored two of the papers above.

Talks: On Better Priors for Policy Search. Invited talk and lab visit, Improbable AI Lab (Pulkit Agrawal), MIT, August 2024.

Reviewer: NeurIPS, ICLR, ICML, RLC

Technical Skills

Computing: Python, JAX, PyTorch, NumPy, Julia, C

Methods: Policy optimization, Bayesian optimization, Gaussian processes, random matrix theory, spectral analysis

Tools: Ray, Weights & Biases, Git

Languages: Arabic (native), English (fluent), French (fluent), German (basic)